NLP.

Experiment - 02.

- **Aim:** Perform bag of words approach (count occurence, normalized count occurence), TF, IDF on data. Create embeddings using Word2Vec. Dataset to be used - https://www.kaggle.com/datasets/CooperUnion/cardataset.

- **Theory:**

(1) **Introduction.**

- Machine learning algorithms cannot directly understand textual data. Therefore, text must be transformed into numerical vectors using feature extraction techniques.
- These techniques capture patterns, frequency and semantic meaning from text.
- In this experiment textual attributes from the Car dataset are processed using three widely used NLP representation methods.

  - Bag of words (BoW)
  - TF - IDF (Term frequency, Inverse document frequency).
  - Word embeddings using Word2Vec.

- The dataset contains structured vehicle information such as make, model, fuel type, vehicle type, transmission and market category. These textual fields are combined to form a corpus where each car description is treated as a document.

This transformation enables tasks such as classification, clustering, reccomendation systems and similarity analysis among vehicles.

(2). Dataset description.

- The car dataset provides detailed specifications of various vehicles including performance metrics and categorical descriptions.

- For NLP representation, the following text attributes are primarily used -
  - Make (eg. - Toyota, BMW)
  - Model (eg - corolla, X5)
  - Engine fuel type (Gasoline, diesel, Electric)
  - Vehicle style (sedan, SUV, convertible)
  - Market category (luxury, performance, hybrid).

- Each record is converted into a single text string representing that vehicle.

- Example document -
  Toyota corolla gasoline Sedan compact automatic.

This collection of douments forms the corpus used for feature extraction.

(3). Bag of words (BoW)

Bag of words is one of the simplest text representation techniques. It constructs a vocabulary of all unique words in the corpus and represents each document as a vector of word frequencies. The order of words is ignored hence the term "bag".

3.1. Count Occurence -

Count occurence measures how many times each word appears in a document

Formula,

$$\text{Count}(w, d) = \text{Frequency of word } w \text{ in document } d.$$

Example document → sedan sedan automatic gasoline.

Vector representation,

- sedan → 2
- automatic → 1
- gasoline → 1.

This produces a sparse vector since most vocabulary words do not appear in a single document.

3.2 Normalized Count Occurence -

Raw counts can be biased by document length. Normalization converts counts into proportions.

Formula →

$$\text{Normalized}(w, d) = \frac{\text{Count}(w, d)}{\sum_i \text{Count}(w_i, d)}$$

Example -

Total words = 4

- sedan → 2/4 = 0.5
- automatic → 1/4 = 0.25
- gasoline → 1/4 = 0.25

Normalized counts allow fair comparison between documents of different lengths.

(4) TF - IDF representation -

- TF-IDF improves Bag of Words by assigning higher weights to informative words and lower weights to commonly occuring words. This helps highlight distinctive vehicle features.

4.1. Term Frequency (TF)

Measures importance of a word within a document

$$TF(w,d) = \frac{Count(w,d)}{Total\ words\ in\ d}$$

4.2. Inverse document Frequency (IDF) -

Measures rarity of a word across the dataset.

$$IDF(w) = \log\left(\frac{N}{1 + DF(w)}\right)$$

Where,

- N = total number of documents.
- DF (w) = number of documents containing the word.

4.3. TF-IDF Score -

$$TF\text{-}IDF(w,d) = TF(w,d) \times IDF(w)$$

* **Advantages:**

- Highlights distinguishing vehicle attributes.
- Reduces noise from frequently occuring generic terms.
- Useful for similarity and search tasks.

(5) **Word embeddings using Word2Vec -**

- Unlike BoW and TF-IDF which produce sparse vectors, Word2Vec generates dense vector representations that capture semantic relationships between words. Words appearing in similar contexts obtain similar vector representations.
- Example -
  - sedan ≈ hatchback.
  - electric ≈ hybrid.
  - sports ≈ performance.

- Architecture -

Word2Vec uses neural network and has two training approaches:

1. Continuous Bag of Words (CBOW) - Predicts the target word using surrounding context words.

2. Skip-Gram - Predicts surrounding words given a target word. Skip gram performs better on small datasets and rare words.

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- Learning objective -

Word2Vec maximises the probability of context words given in a target word.

$$\max \Sigma \log P(\text{context} \mid \text{target})$$

This results in meaningful vector spaces where semantic similarity can be measured using cosine similarity.

* Benefits -
  - Captures semantic similarity between vehicle categories.
  - Enables clustering of similar cars.
  - Improves recommendation systems.
  - Provides dense low dimensional features for ML models.

(6). Applications on Car Dataset -

- Vehicle similarity search.
- Market segmentation.
- Recommendation of similar cars.
- Classification (luxury vs economy).
- Trend analysis of fuel types and vehicle styles.

(7) Comparative analysis -

	Technique	Vector Type	Key Advantage	Limitation
1.	Bag of words	Sparse	Simple, interpretable	Ignores semantic.
2.	TF IDF	Weighted sparse	Highlights terms	High dimensional
3.	Word 2Vec	Dense	Captures semantic	Training required.

* CONCLUSION: Thus we successfully performed bag of words on a car dataset.